MILITARY SPECIFICATION

HOOKS, CARGO (DROP FORGED)

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. - This specification covers commercially available hooks for general cargo hoisting.

1.2 Classification. - Cargo hooks shall be furnished in the following types, as specified (see 6.2):

Type I - Portland
Type II - Liverpool
Type III - Western
Type IV - Seattle
Type VII - Seattle (with yokes)

2. APPLICABLE DOCUMENTS

2.1 The following document of the issue in effect on date of invitation for bids or request for proposal, forms a part of the specification to the extent specified herein:

SPECIFICATIONS

MILITARY
MIL-H-15424 - Hand Tools; Packaging of.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal, shall apply.

AMERICAN IRON AND STEEL INSTITUTE (AISI)
Steel Products Manual.

(Application for copies should be addressed to the American Iron and Steel Institute, 350 5th Avenue, New York 1, New York.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
B-144 - High-Leaded Tin Bronze Sand Castings.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Materials. -

3.1.1 General requirements. - Cargo hooks shall be drop-forged, without welding, from 1020 to 1030 normalized steel in accordance with AISI Steel Products Manual.

3.1.2 Yokes, shackles and swivel eyes. - Yokes, shackles and swivel eyes shall be forged from 1020 to 1030 normalized steel in accordance with AISI Steel Products Manual.

3.1.3 Washers. - Washers for swivel bearings shall be made of alloy 3C bronze in accordance with ASTM B-144.

3.2 Design. -

3.2.1 Type I, Portland. - The type I hook shall be designed plain and shall conform to the requirements of figure 1 and table I.



Figure 1 - Type I Portland hook.

Table I - Type, size, capacity, length and weight.

Type	Size number	Capacity tons (2240 pounds)	Length of hook (inches)	Approximate weight (pounds)	Length of assembly (inches)	Packed per case
I	22	5	13	14-1/2	--	12
II	1	2-1/2	10-1/2	12-3/4	--	16
II	2	5	13	23	--	8
III	32	5	13	35	--	4
IV	42	5	13	44	--	4
VII	42	5	13	--	35-3/4	4
VII	43	10	13	--	35-3/4	4

3.2.2 Type II, Liverpool. - The type II hook shall be designed with a single swivel and shall conform to the requirements of figure 2 and table I.



Figure 2 - Type II Liverpool hook.

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3.2.3 Type III, Western. - The type III hook shall be designed with a ring and two free swivels and shall conform to the requirements of figure 3 and table I.

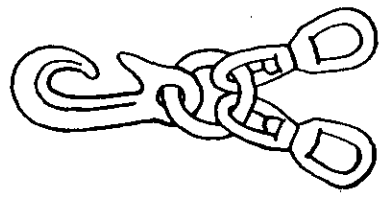


Figure 3 - Type III Western hook.

3.2.4 Types IV and VII, Seattle. -

3.2.4.1 Type IV. - The type IV hook shall be designed with a swivel, a ring and two free swivels and shall conform to the requirements of figure 4 and table I.

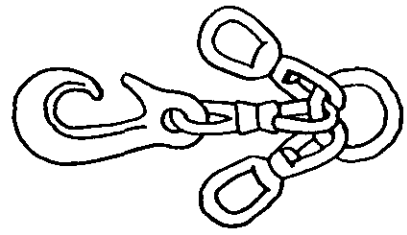


Figure 4 - Type IV Seattle hook.

3.2.4.2 Type VII. - The ype VII hook shall be designed with a yoke, a shackle, two swivels and a pin. The pin shall be moly-chrome hexagon head and nut steel pin with corners rounded. The hook shall conform to the requirements of figure 5 and table I.

3.3 Marking. - Each hook shall be permanently marked with the working load in pounds, and the manufacturer's name or trademark.

3.4 Workmanship. - All parts shall be properly shaped and free from fins, cracks, nicks, flaws, seams and other injurious defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. -

4.2.1 Lot. - A lot shall consist of all hooks of the same size and type offered for delivery at one time.

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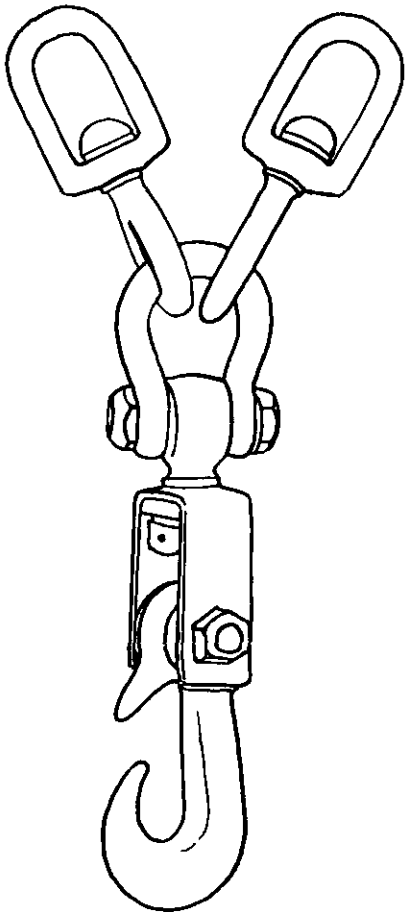


Figure 5 - Type VII Seattle hook (with yokes).

4.2.2 Sampling for examination. - Sample hooks shall be selected at random from each lot in accordance with table II for the examination specified in 4.3. If the number of defective hooks exceeds the acceptance number for the sample, this shall be cause for rejection of the lot represented by the sample. *Table II shall be applied separately to major or minor defects* for purposes of lot acceptance and rejection.

4.2.3 Sampling for proof test. - Sample hooks shall be selected from each lot in accordance with table III for the test of 4.4.1. If any sample hook fails this test, this shall be cause for rejection of the entire lot represented by the sample.

4.2.4 Sampling for ultimate strength test. - Sample hooks shall be selected from each lot in accordance with table III for the test specified in 4.4.2. If any hook fails this test, this shall be cause for rejection of the entire lot.

4.3 Examination. - Each sample hook selected in accordance with 4.2.2 shall be examined to verify conformance to all the requirements of this specification not involving tests. Defects shall be classified in accordance with table IV.

4.4 Tests. -

4.4.1 Proof test. - Each sample hook selected in accordance with 4.2.3 shall be subjected to a proof test of twice the rated capacity specified in table I.

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Table II - Sampling for examination.

Lot size (number of hooks)	Sample size (number of hooks)	Defective hooks	
		Acceptance number (defectives)	Rejection number (defectives)
2 to 8	5	0	1
9 to 15	7	0	1
16 to 25	10	0	1
26 to 40	15	0	1
41 to 65	25	0	1
66 to 110	35	1	2
111 to 180	50	2	3
181 to 300	75	3	4
301 to 500	110	4	5
501 to 800	150	5	6

Table III - Sampling for tests.

Lot size	Sample size	
	Proof test	Ultimate strength test
Up to 15	5	1
16 to 40	7	2
41 to 110	10	3
111 to 180	15	5
181 to 300	25	7
301 to 500	35	10
501 to 800	50	15

Table IV - Classification of Defects

Categories	Defects
Critical:	None defined.
Major:	
101	Type and size not as specified.
102	Length not as specified.
103	Incomplete; components, when applicable, missing or improperly positioned or attached.
104	Not shaped in accordance with figures in specification.
105	Material not as specified; not forged, not free from fins, cracks, flaws or seams.
106	Eye not parallel with hook, not properly positioned.
107	Surfaces not free from burrs, sharp edges, not smooth, evidence of rust or corrosion.
108	Marking of capacity of hook not as required.

4.4.2 Minimum ultimate strength test. - Each sample hook selected in accordance with 4.2.4 shall be tested to four times the rated capacity specified in table I.

4.4.3 Possible test failures. -

(a) Proof test load. -
(1) Evidence of deformation, distortion or permanent set, signs of incipient cracks in hook or any component parts of hook assembly after subjecting hook to twice the rated capacity.

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(b) Minimum ultimate strength test. -
(1) Ultimate strength of hook or hook assembly less than four times the rated capacity.

4.5 Examination for preparation of delivery. - Examination of the packaging and packing shall be made for conformance with the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. - Hooks shall be preserved and packaged level A or C, packed level A, B or C and marked in accordance with MIL-H-15424 as specified (see 6.2). Quantities shall be packed in accordance with table I.

6. NOTES

6.1 Intended use. - Cargo hooks covered by this specification are intended for hoisting cargo.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type hook required (see 1.2).
(c) Selection of applicable levels of preservation, packaging and packing required (see 5.1).

Notice. - When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Custodians:
Army - WC
Navy - Ships
Air Force - MOA

Preparing activity:
Navy - Ships
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